
Lecture Notes

on Basic Descriptive Statistics

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1 Types of Data

- (1) Data on a ratio scale
- (2) Data on an Interval scale
- (3) Data on ordinal scale
- (4) Data on Nominal scale
- (5) Continuous vs. Discrete data

1.1 Definitions and examples

1.1.1 Ratio scale

A measurement scale that has constant interval size and a **true zero point**.

☛ **Examples:** age, height, distance etc.

↳ a true zero point means none
as 0 cm height = no height
(does not exist)

1.1.2 Interval scale

A measurement scale with an interval size but have **no true zero point**.

☛ **Examples:** temperature, date on a calendar, time on a watch etc.

↳ 0°C or 0°F means some value i.e, not none.

1.1.3 Ordinal scale

In this scale data can be ordered according to some measurement.

☛ **Examples:** education level (primary, secondary or post secondary), Income (low, middle, and high) etc.

1.1.4 Nominal scale

In this scale a measurement is classified by an attribute rather than by a quantity measurement.

☛ **Examples:** grade scale (A,B,C,D), gender (Female, Male), blood group (A, B, AB, O), species (bird, mammal) etc.

1.1.5 Continuous and Discrete data

Continuous data: This kind of data can take any value within a given range. [0, 3]

☛ **Examples:** height, temperature etc.

Discrete data: This kind of data consists of distinct, separate values that can be counted. 0, 1, 2, 3

☛ **Examples:** number of students in a classroom, number of cars in a parking lot, books on a shelf etc.

1.2 Tools used to describe and summarize data

(1) Measures of central tendency

(2) Measures of dispersion

1.2.1 Measures of central tendency

These summarize a data set by a single value point mostly center value.

☛ **Examples of measures of central tendency:**

Arithmetic mean (mean): Let $\{x_1, x_2, \dots, x_n\}$ denotes a data set with n data points. Then the arithmetic mean of these data points is defined as

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}.$$

☑ **Example:** Suppose the marks of five students are $\{50, 60, 70, 80, 90\}$. Then, the arithmetic mean or mean is given by

$$\bar{x} = \frac{50 + 60 + 70 + 80 + 90}{5} = 70.$$

Median: The median is the middle value of an **ordered dataset**.

- If the number of data points n is odd, the median is the value at position $\frac{(n+1)}{2}$.
- If n is even, the median is the average of the values at positions $\frac{n}{2}$ and $(\frac{n}{2}) + 1$.

☑ **Example:** Consider the dataset $\{3, 13, 2, 34, 11, 26, 47\}$ ($n = 7$).

Ordered set: $\{2, 3, 11, 13, 26, 34, 47\}$

Here $n = 7$ is odd, so median is the average of the

$(\frac{7+1}{2}) = 4^{\text{th}}$ place data pt i.e, 13.

✍ **Example:** Consider the dataset $\{3, 13, 2, 34, 11, 17, 27, 47\}$ ($n = 8$).

Ordered set: $\{2, 3, 11, 13, 17, 27, 34, 47\}$.

Now, $\frac{n}{2} = \frac{8}{2} = 4$, & $\frac{n}{2} + 1 = 4 + 1 = 5$.

So, the median = $\frac{13+17}{2} = 15$.

Difference between mean and median: Consider the data set: $\{0, 0, 0, 1, 1, 2, 10, 10\}$. Find the mean and median.

$$\text{Mean} = \bar{x} = \frac{0+0+0+1+1+2+10+10}{8} = \frac{24}{8} = 3$$

Here $n=8$, so the median is given by the average of
 $\frac{n}{2} = \frac{8}{2} = 4^{\text{th}}$ place data pt = 1 & $\frac{n}{2} + 1 = \frac{8}{2} + 1 = 5^{\text{th}}$ place data pt = 1,
 i.e., $\frac{1+1}{2} = 1$.

Mode: The mode of a dataset is the value (or values) that occur most frequently. A dataset may have:

- **One mode (unimodal):** e.g., $\{2, 4, 4, 6, 7\}$ has mode 4.
- **Two modes (bimodal):** e.g., $\{1, 2, 2, 3, 3, 4\}$ has modes 2 and 3.
- **More than two modes (multimodal):** e.g., $\{5, 6, 6, 7, 7, 8, 8\}$ has modes 6, 7, 8.
- **No mode:** if all values occur with the same frequency.

Midrange: The midrange of a dataset is the value halfway between the minimum and maximum data values. It is calculated as

$$\text{Midrange} = \frac{\text{Minimum value} + \text{Maximum value}}{2}.$$

✍ **Example:** For the dataset $\{3, 7, 10, 15, 18\}$,

$$\text{Midrange} = \frac{3+18}{2} = 10.5.$$

Geometric Mean: Let $\{x_1, x_2, \dots, x_n\}$ be a dataset of n positive numbers. The geometric mean is defined as the n^{th} root of the product of those n points, and is given by

Σ : notation of sum

Π : notation of product

If we say $a \in \mathbb{R}$ means a is real number. $n\sqrt{}$

$$\text{GM} = (x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}.$$

$$\begin{aligned} * \sqrt[2]{4} &= \sqrt{4} = (2^2)^{\frac{1}{2}} \\ &= 2^{2 \times \frac{1}{2}} = 2^1 = 2 \end{aligned}$$

$$* \sqrt[3]{8} = (2^3)^{\frac{1}{3}} = 2^{3 \times \frac{1}{3}} = 2^1 = 2$$

$$* 2^0 = 1$$

✎ **Example:** For the dataset $\{4, 16\}$,

$$GM = \sqrt{(4 \times 16)} = 8.$$

✎ For the dataset $\{2, 8, 18\}$ with $n = 3$,

$$GM = \sqrt[3]{2 \times 8 \times 18} = 6.60.$$

Harmonic Mean: Let $\{x_1, x_2, \dots, x_n\}$ be a dataset of n positive numbers. The harmonic mean is defined as the reciprocal of the arithmetic of the reciprocal of the data points, and is given by

$$\frac{1}{x_1}, \frac{1}{x_2}, \dots, \frac{1}{x_n}$$

$$HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}.$$

find the arithmetic mean $\rightarrow$ then take the reciprocal.

✎ **Example:** For the dataset $\{4, 8, 16\}$ with $n = 3$,

$$HM = \frac{3}{\frac{1}{4} + \frac{1}{8} + \frac{1}{16}}$$

$$1) \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$$

$$2) \frac{\frac{1}{4} + \frac{1}{8} + \frac{1}{16}}{3}$$

$$\frac{\frac{a}{b} + \frac{c}{d}}{= \frac{ad + bc}{bd}}$$

$$\begin{aligned} \frac{\frac{2}{3} + \frac{5}{2}}{=} &= \frac{(2 \times 2) + (5 \times 3)}{6} \\ &= \frac{4 + 15}{6} \\ &= \frac{19}{6} \end{aligned}$$

$$\begin{aligned} \frac{\frac{1}{2} + \frac{1}{3}}{=} &= \frac{3 + 2}{6} \\ &= \frac{5}{6} \end{aligned}$$

1.2.2 Measures of dispersion

These describe the spread of data points around the central tendency.

✎ **Examples of measures of dispersion:**

Range: The range of a dataset is the difference between the maximum and minimum values.

$$\text{Range} = \text{Maximum value} - \text{Minimum value}.$$

✎ **Example:** For the dataset $\{5, 8, 12, 20, 25\}$,

$$\begin{aligned} \text{Range} &= 25 - 5 \\ &= 20. \end{aligned}$$

Variance: The variance of a dataset measures the average of the squared deviations from the mean. Since most datasets are considered as samples, we compute variance using $(n - 1)$ in the denominator (Bessel's correction):

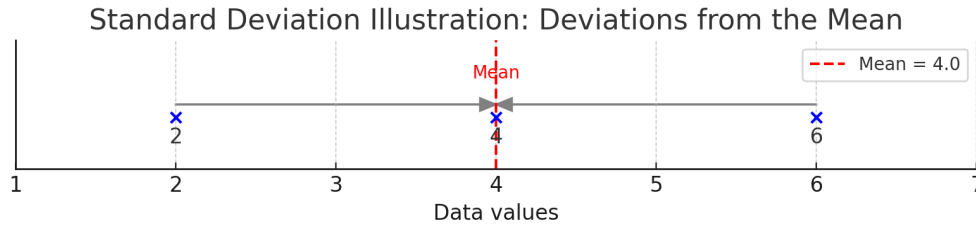


Figure 1: Illustration of standard deviation: dataset $\{2, 4, 6\}$ with the mean marked in red and arrows showing deviations from the mean.

$$\begin{aligned}
 & (x_1 - \bar{x}), (x_2 - \bar{x}), \dots, (x_n - \bar{x}) \\
 & \hookrightarrow (x_1 - \bar{x})^2, (x_2 - \bar{x})^2, \dots, (x_n - \bar{x})^2 \rightarrow \sum_{i=1}^n (x_i - \bar{x})^2 \rightarrow \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{(n-1)}
 \end{aligned}$$

$$\sigma^2 \text{ (or } s^2) = \frac{1}{(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Sigma (σ)

✎ **Example:** For the dataset $\{2, 4, 6\}$ with $n = 3$, Here, $x_1 = 2, x_2 = 4, x_3 = 6$.

$$\bar{x} = \frac{2 + 4 + 6}{3} = 4.$$

$$s^2 = \frac{8}{(3-1)} = \frac{8}{2} = 4.$$

$$1) \bar{x} = 4$$

$$2) (x_1 - \bar{x}) = (2 - 4) = -2,$$

$$(x_2 - \bar{x}) = (4 - 4) = 0,$$

$$(x_3 - \bar{x}) = (6 - 4) = 2,$$

$$3) (-2)^2 + 0^2 + (2)^2 = 8.$$

Standard Deviation: The standard deviation is a measure of dispersion that indicates how much the data values deviate, on average, from the mean. It is defined as the **square root of the variance**:

$$s = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2}.$$

$$\sqrt{16} = \pm 4$$

✎ **Example:** For the dataset $\{2, 4, 6\}$ with $n = 3$,

$$\bar{x} = 4, \quad s^2 = 4, \quad s = \sqrt{4} = 2.$$

Coefficient of Variation (CV): The coefficient of variation is a standardized measure of dispersion that expresses the **standard deviation as a percentage of the mean**. It is defined as

$$CV = \frac{s}{\bar{x}} \times 100\%,$$

$$\frac{2}{5} \times 100\% = (2 \times 20)\% = 40\%$$

where s is the standard deviation and $\bar{x}$ is the arithmetic mean.

✎ **Example:** For the dataset $\{2, 4, 6\}$,

$$\bar{x} = 4, \quad s = 2, \quad CV = \frac{2}{4} \times 100\% = 50\%.$$